OCCUQ: Exploring Efficient Uncertainty Quantification for 3D Occupancy Prediction

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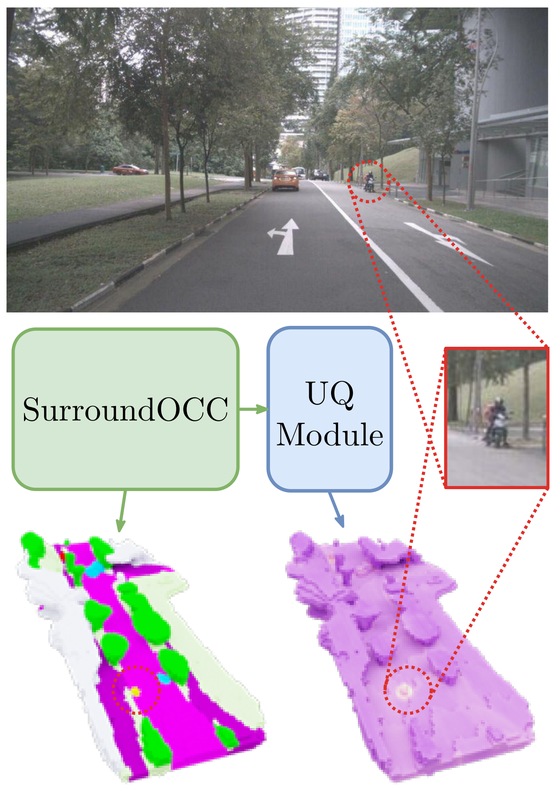
Introduction

- **Quantifying Uncertainty** is vital for **reliable** perception and enabling **safety** mechanisms in **autonomous systems**.
- **Uncertainty Estimation** is well-studied in literature, but current methods often come with **high computational costs**.
- **Efficient Uncertainty Estimation** has not yet been investigated for **3D Occupancy Prediction**
- We propose an efficient adaptation of a **feature-density-based approach**.
- We evaluate the approach's ability to estimate **epistemic uncertainty** for **out-of-distribution detection** on **scene** and **region level**.
- **Our approach** provides **reliable uncertainty estimates** with **minimal computational overhead**.



Method

We use **feature density** as an **uncertainty measure** following the **DDU methodology** [1]. To ensure that the **feature density** correlates with the **uncertainty**, a **well-regularized feature space** is necessary. We leverage **SurroundOcc** [2] as occupancy predictor, which can be expressed as $V(I) = g \circ f_\theta(I)$ with the backbone $f_\theta: I \rightarrow \mathcal{H}$, that maps input images I to a feature representation $f_\theta(I) \in \mathcal{H}$, and an output function g that maps $f_\theta(I)$ to the label space. To make the model **distance aware**, we require the **bi-Lipschitz constraint** on f_θ:

$$C_L D_{input}(x_1, x_2) \leq D_{feat}(f_\theta(x_1), f_\theta(x_2)) \leq C_U D_{input}(x_1, x_2)$$

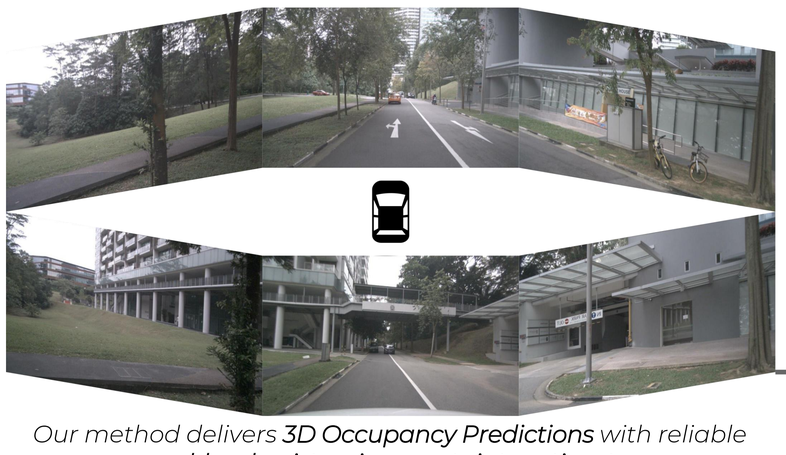
for all inputs x_1 and x_2, where D_{input} and D_{feat} denote distance metrics for the input and feature space, and C_L and C_U the lower and upper Lipschitz constants. This ensures **sensitivity** to input changes while providing **smoothness** against minor variations.

- We use **Residual Connections** and apply **Spectral Normalization (SN) in an UQ Module** to satisfy the Lipschitz constraints (**Smoothness** and **Sensitivity**).
- A GMM $q(y|z)$ **is fitted on the normalized features** using the **training set**.
- At test time: **Epistemic uncertainty** is the marginal **likelihood** under the **fitted GMM**:

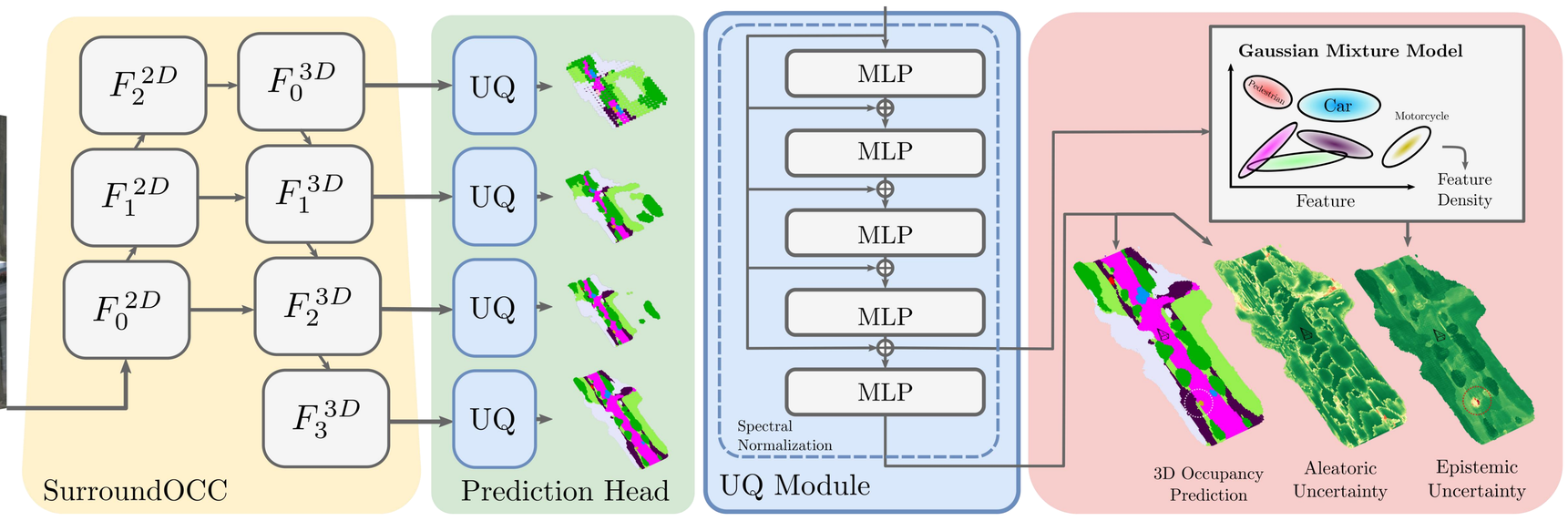
$$q(z) = \sum_y q(z|y)q(y)$$

- High feature-space density = Low epistemic uncertainty
- Low feature-space density = High epistemic uncertainty

Overview

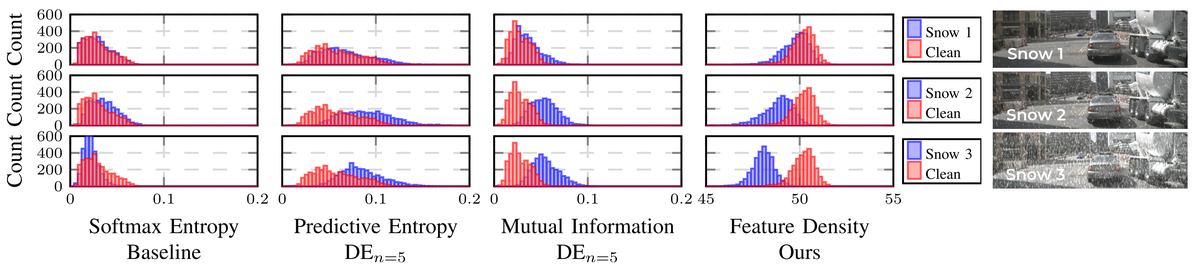


*Our method delivers **3D Occupancy Predictions** with reliable **voxel-level epistemic uncertainty estimates**.*



Experiments & Results

We evaluate **our approach** in capturing **epistemic uncertainty** using the **nuScenes** and **MultiCorrupt** [3] dataset. Due to the unavailability of voxel-level OoD data, we establish an **OoD detection** framework at **scene** and **region level**. In this context, our approach outperforms the **baseline** model, **Monte Carlo Dropout (MCD)** and **Deep Ensembles (DE)**.



*Corruption **"Snow"** impacts the **feature density distribution** of **our proposed method**, leading to a **better separation** between **ID** and **OoD samples**.*

Method	nuScenes IoU↑	nuScenes mIoU↑	MultiCorrupt IoU↑	MultiCorrupt mIoU↑	Scene Corruptions mAUROC↑	Scene Corruptions mFPR95↓	Region Corruptions mAUROC↑	Region Corruptions mFPR95↓	Params↓	Time(s)↓	Memory↓
$MCD_{n=5}$	0.331	0.209	0.242	0.137	63.03	87.44	51.21	94.45	**180.07M**	2.11	**7.1GB**
$DE_{n=3}$	0.338	0.218	0.246	0.145	68.46	76.64	54.26	94.16	540.22M	1.26	21.3GB
$DE_{n=5}$	**0.341**	**0.221**	**0.247**	**0.148**	69.58	74.89	55.26	94.23	900.37M	2.11	35.5GB
Entropy	0.328	0.208	0.242	0,141	54.12	95.06	50.01	94.32	**180.51M**	**0.42**	**7.1GB**
Max. Softmax	0.328	0.208	0.242	0,141	55.42	87.71	50.22	95.65	**180.51M**	**0.42**	**7.1GB**
Ours	0.329	0.210	0.242	0.141	**80.17**	**56.39**	**57.81**	**93.60**	180.51M	0.45	7.1GB

*Our method demonstrates **superior performance** in terms of **epistemic uncertainty quantification** while maintaining the properties of the **base model**.*

Ablation Studies

- Prediction head design (**spectral normalization, skip connections**) is key for **distance awareness**
- **Skip connections** are essential for **sensitive features**

Layers	Skip	mAUROC↑	mFPR95↓	AP↑	Params↓
3		0.770	0.618	0.760	180.336M
5		0.674	0.761	0.678	180.511M
3	✓	0.758	0.659	0.747	180.336M
5	✓	**0.802**	**0.564**	**0.794**	180.511M

OoD detection performance for different UQ module configs.

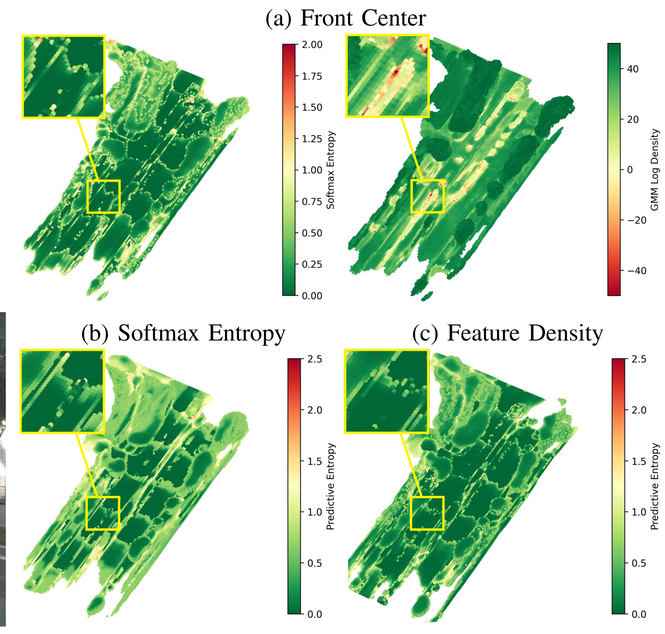
- Impact of **output resolution** on OoD detection performance
- **Finer resolutions** substantially improve ID/OoD separation
- **Finer voxels** lead to a **more accurate GMM**

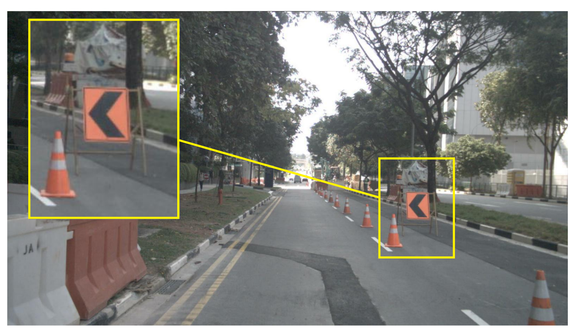
Resolution	Dim.	mAUROC↑	mFPR95↓	Params↓
20x20x2	256	58.00	90.81	1,118,464
50x50x4	128	72.01	74.05	280,704
100x100x8	64	75.59	69.93	69,649
200x200x16	32	**80.17**	**56.39**	**17,952**

OoD detection performance at different output scales.

Visualization

- Cam (a) shows a **challenging scene**
- Baselines (b, d, e) miss uncertainty
- Our approach (c) correctly highlights **high uncertainty** around **rare objects**





(a) Front Center (d) $DE_{n=5}$ PE (e) $MCD_{n=5}$ PE

Code & Dataset

- GMM Training and GMM GPU Inference
- Trained weights ready to download
- Dataset for OoD Detection
- Evaluation & Visualization tools
- *github.com/ika-rwth-aachen/OCCUQ*

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Conclusion

- Proposed an **Uncertainty Quantification method for 3D Occupancy Prediction**.
- Adapted the **DDU method** for **voxel-wise epistemic uncertainty estimation**.
- Method **effectively identifies OoD data (scene/region level)** while **maintaining prediction performance** of unmodified baseline model.
- Lays **groundwork for safe and reliable 3D Occupancy Prediction**.
- Future Work: **Need for voxel-level OoD datasets** for better evaluation.

[1] **DDU:** Deep Deterministic Uncertainty: A New Simple Baseline, Mukhoti et al., CVPR 2023; [2] **SurroundOcc:** Multi-camera 3D Occupancy Prediction for Autonomous Driving, Wei et al. ICCV 2023;
[3] **MultiCorrupt:** A Multi-Modal Robustness Dataset and Benchmark of LiDAR-Camera Fusion for 3D Object Detection, Beemelmanns et al. IV2024